

RMIT University
School of Computing Technologies
Practical Data Science with Python
Assignment 1: Data Cleaning and Summarising
Due: 23:59 on the 19th of April 2026
This assignment is worth 25% of your overall mark.

Introduction

In this assignment, you will examine a (set of) data file(s) and carry out the first steps of the data science process, including the cleaning and exploring of data. You will need to develop and implement appropriate steps, in Jupyter notebook (in the Anaconda version specified in the canvas announcement), to load a data file into memory, clean, process, and analyse it. This assignment is intended to give you practical experience with the typical first steps of the data science process.

The “Practical Data Science” Canvas contains further announcements and a discussion board for this assignment. Please be sure to check these on a regular basis – it is your responsibility to stay informed with regards to any announcements or changes. Login through <https://rmit.instructure.com/>.

Where to Develop Your Code

You are encouraged to develop and test your code in the following environments: **Jupyter Notebook on Lab PCs**. (Or please use the Anaconda version as specified in the course announcement)

Jupyter Notebook on Lab PCs

On Lab Computer, you can find Jupyter Notebook via:

Start → All Programs → Jupyter Notebook - Anaconda

Then,

- Select New → Python 3
- The new created ‘*.ipynb’ is created at the following location:
 - C:\Users\sXXXXXXX

- where sXXXXXXX should be replaced with a string consisting of the letter “s” followed by your student number.

Plagiarism

RMIT University takes plagiarism very seriously. All assignments will be checked with plagiarism-detection software; any student found to have plagiarised will be subject to disciplinary action as described in the course guide. Plagiarism includes submitting code that is not your own or submitting text that is not your own. Allowing others to copy your work is also plagiarism. All plagiarism will be penalised; there are no exceptions and no excuses. For further information, please see the *Academic Integrity* information at <http://www1.rmit.edu.au/academicintegrity>.

Turnitin will be used for Plagiarism Review for this assignment in Canvas.

General Requirements

This section contains information about the general requirements that your assignment must meet. *Please read all requirements carefully before you start.*

- You *must* do the assignment in **Jupiter Notebook** that are available in Anaconda.
- Parts of this assignment will include a written report; this *must* be in **PDF** format.
- Please ensure that your submission follows the file naming rules specified in the tasks below. File names are case sensitive, i.e. if it is specified that the file name is gryphon, then that is exactly the file name you should submit; Gryphon, GRYPHON, griffin, and anything else but gryphon will be rejected.

Task 1: Data Preparation (8 marks)

Download `victorian_road_crash_data.csv` from Canvas under the Assignments → Assignment 1 section of the course Canvas. It provides a set of attributes for each road crash that has occurred within Victoria.

Being a careful data scientist, you know that it is vital to carefully check any available data before starting to analyse it. Your task is to prepare the provided data for analysis. You will start by loading the CSV data from the file (using appropriate pandas functions) and checking whether the loaded data is equivalent to the data in the source CSV file. Next, apply the data cleaning techniques covered in the lectures to resolve issues/errors thoroughly and completely.

Once the data is cleaned, store the refined version as a CSV file, e.g. (file can be named as 'cleaned version.csv').

Task 2: Data Exploration (10 marks)

Using the **cleaned** data, answer the questions below.

Q1: How do accident frequencies vary by the **Day of the Week** and **Time of Day** in the year of 2023 and how this relates to lighting (`LIGHT_CONDITION`)? What insights can be gained from this exploration?

Q2: What is the relationship between the `SPEED_ZONE` and the number of `INJ_OR_FATAL` victims? Visualize the distribution of speed zones for accidents labeled as "Serious injury accident" versus "Other injury accident".

Q3: Analyse crash distribution by geographic area and relate it to at least one risk factor.

Required:

- identify and visualise the **top 5 areas** (choose one dimension):
 - `LGA_NAME` (Local Government Area), or
 - `DTP_REGION`
- for those top 5 areas, compare at least **two** of:
 - severity (fatal/serious injury rates)
 - `SPEED_ZONE` distribution
 - `ROAD_GEOMETRY` / `ROAD_GEOMETRY` types
 - involvement of vulnerable road users (`PEDESTRIAN`, `BICYCLIST`, `MOTORCYCLIST`)

Interpretation required: tell a short story (e.g., what differs across areas, and what might explain it based on the variables you analysed).

Note, each visualization (graph) should be complete and informative in itself and should be clear for readers to read and obtain information.

Task 3: Report (7 marks)

Write your report and save it as **report.pdf**:

- PDF only
- max **6 pages**, single column (including everything, e.g. figures, references etc)

- font size **10–12**
- Clarity, structure/ grammar, and writing quality matter
- Remember to clearly cite any sources (including books, research papers, course notes, etc.) that you referred to while designing aspects of your programs.
- Create a heading called “Data Preparation” in your report.
 - Provide a brief explanation of how you addressed the task. For the steps of dealing with the potential issues/errors, please create a sub-section for each type of errors you dealt with (e.g. typos, extra whitespaces, sanity checks for impossible values, and missing values etc), and also explain and justify how you dealt with each kind of errors.
- Create a heading called “Data Exploration” in your report.
 - For each numbered step in Task 2 above, create a sub-section Q1, Q2, Q3. Please explain and justify how you explore the data as required properly.

What to Submit, When, and How

The assignment is due at 23:59 on the 19th of April 2026.

Assignments submitted after this time will be subject to standard late submission penalties.

You need to submit the following files:

- Notebook file containing your python commands for Task 1 and Task2, ‘assignment1.ipynb’. **Please use the provided solution template to organise your solutions:** *assignment1 TEMPLATE.ipynb*
- # For the notebook files, please make sure to clean them and remove any unnecessary lines of code (cells). Follow these steps before submission:
 1. Main menu → Kernel → Restart & Run All
 2. Wait till you see the output displayed properly. You should see all the dataprinted and graphs displayed.
- Your report.pdf file: **at most 6 (in single column format) pages (including figures and references) with a font size between 10 and 12 points.** Penalties will apply if the report does not satisfy the requirement.

They must be submitted as ONE single zip file, named as your student number (for example, 1234567.zip if your student ID is s1234567). The zip file must be submitted in Canvas:

Assignments/Assignment 1.

Please do NOT submit other unnecessary files.